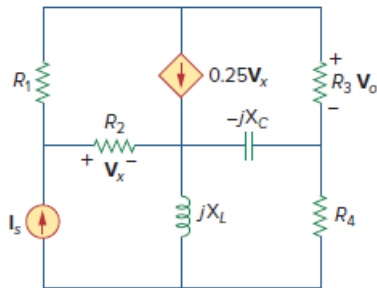


Problem

Using Fig. 10.127, design a problem to help other students better understand performing AC analysis with *PSpice* or *MultiSim*.

Figure 10.127:



Step-by-step solution

Step 1 of 4

Refer to Figure 10.127 in the textbook.

Consider the value of the angular frequency.

$$\omega = 1 \text{ rad/s}$$

Assume the values of the elements in the circuit are,

$$R_1 = 1 \, \Omega$$

$$R_2 = 1 \, \Omega$$

$$R_3 = 1 \, \Omega$$

$$R_4 = 1 \, \Omega$$

$$L_1 = 1 \text{ H}$$

$$C_1 = 1 \text{ F}$$

The amplitude of the voltage source as 1 V .

Calculate the frequency, f .

$$\begin{aligned} f &= \frac{\omega}{2\pi} \\ &= \frac{1}{2\pi} \\ &= 0.15915 \text{ Hz} \end{aligned}$$

Step 2 of 4

Draw the circuit in PSPICE schematics.

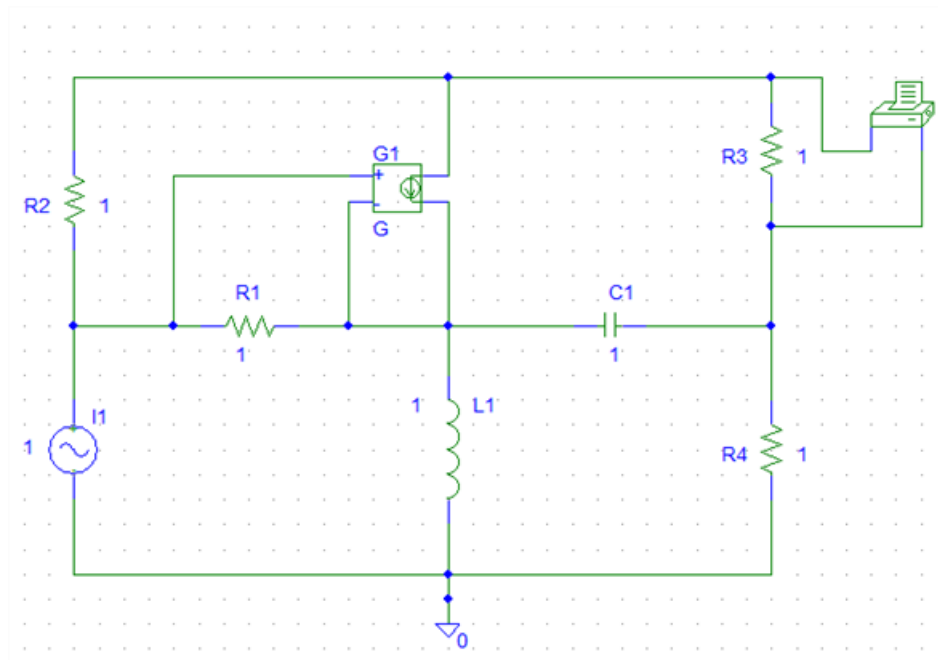


Figure 1

Step 3 of 4

The magnitude and phase of V_o are required. So, set the attributes of V PRINT1 to AC = yes, MES = yes, PHASE = yes.

Go to 'Analysis' menu open setup / AC sweep and enter total points as 1, start frequency as 0.15915 and Final frequency as 0.15915.

Save the Schematic and simulate it by selecting analysis/simulate. The output file includes the source frequency in addition to the attributes checked for the pseudo component VPRINT as follows:

The output of the circuit is,

FREQ VM(\$N_0003,\$N_0004) VP(\$N_0003,\$N_0004)

1.592E-01 4.282E-01 -1.045E+02

Therefore, the value of the voltage V_o is $0.482 \angle -104.5^\circ \text{ V}$.

Step 4 of 4

Go to 'Analysis' menu open setup / AC sweep and enter total pts as 1, start frequency as 0.15915 and Final frequency as 1 kHz.

The outputs of the circuit are,

FREQ VM(\$N_0003,\$N_0004) VP(\$N_0003,\$N_0004)

1.592E-01 4.282E-01 -1.045E+02

1.113E+02 2.308E-01 -1.201E+02

2.223E+02 2.308E-01 -1.201E+02

3.334E+02 2.308E-01 -1.200E+02

4.445E+02 2.308E-01 -1.200E+02

5.556E+02 2.308E-01 -1.200E+02

6.667E+02 2.308E-01 -1.200E+02

7.778E+02 2.308E-01 -1.200E+02

8.889E+02 2.308E-01 -1.200E+02

1.000E+03 2.308E-01 -1.200E+02

Therefore, the value of the voltage V_o is $0.482 \angle -104.5^\circ \text{ V}$ at the frequency of

0.15915 Hz and $2.308 \angle -120^\circ \text{ V}$ at the remaining all frequencies and the output changes by changing different values.